

Filtering II - frequency response and stability

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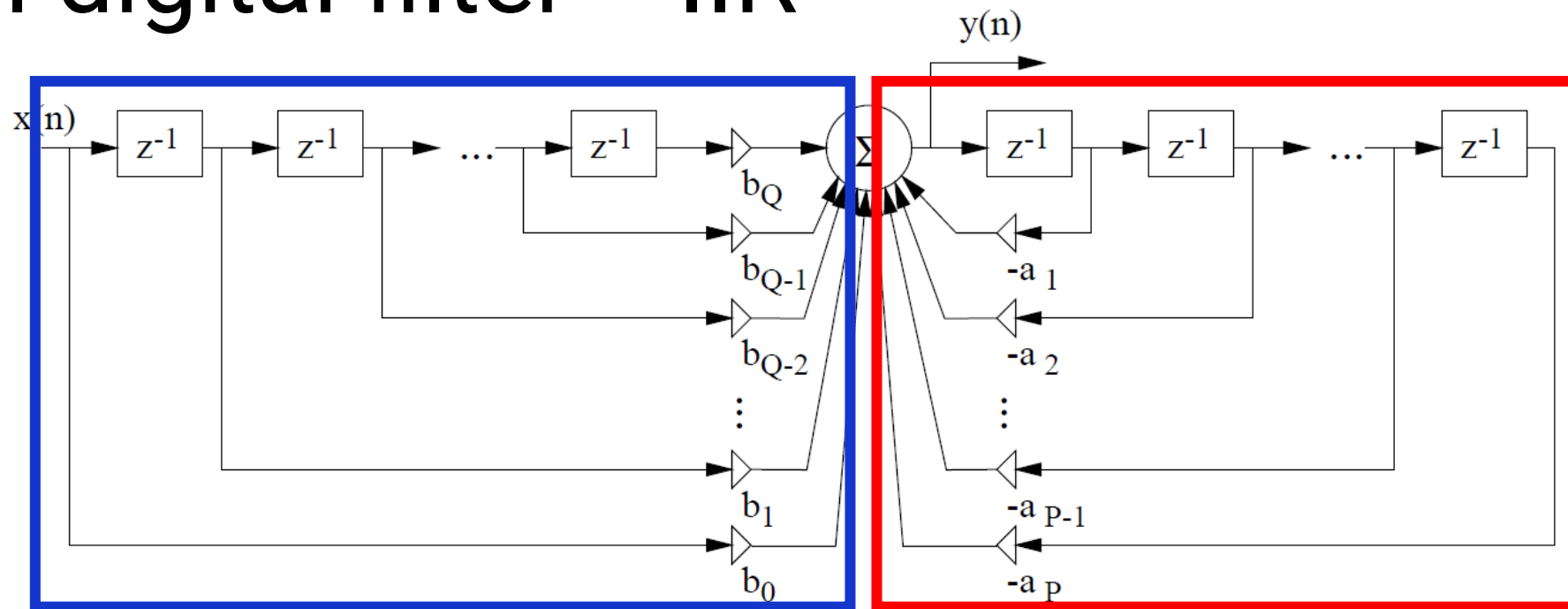
Thanks Petr Pálka (BP 2021/22) for great visualization functions in Python

Please open Python notebook **filtering 2**

Agenda

- **Example of an IIR filter**
- Time domain vs. frequency domain
- z-transform and transfer function
- Frequency response
- Factorizing the transfer function
- Stability
- Summary

General digital filter - IIR



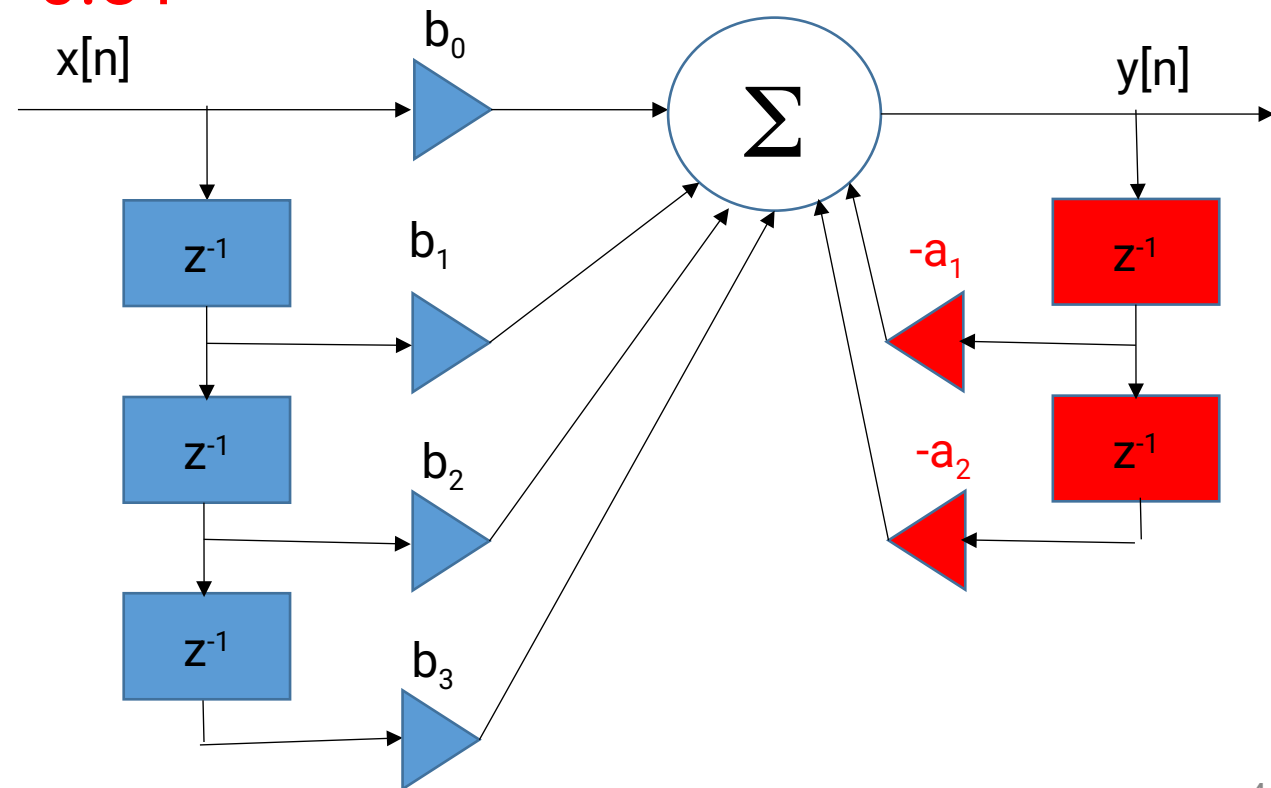
- Scheme
- Difference equation (**diferenční rovnice**)

$$y[n] = b_0x[n] + b_1x[n-1] + \dots + b_Qx[n-Q] - a_1y[n-1] - \dots - a_Py[n-P]$$

$$y[n] = \sum_{k=0}^Q b_k x[n-k] - \sum_{k=1}^P a_k y[n-k]$$

Example of a real filter

- $b_0 = 1, b_1 = -1, b_2 = 1, b_3 = -1$
- $a_1 = -1.2723, a_2 = 0.81$



It can really be programmed !

```
float filter (float xn) {  
    float b0 = 1, b1 = -1, b2 = 1, b3 = -1,  
          a1 = -1.2723, a2 = 0.81;  
    static float xn1 = 0.0, xn2 = 0.0, xn3 = 0.0;  
    static float yn1 = 0.0, yn2 = 0.0,  
    float yn;  
    yn = b0 * xn + b1 * xn1 + b2 * xn2 + b3 * xn3  
        - a1 * yn1 - a2 * yn2;  
    xn3 = xn2; xn2 = xn1; xn1 = xn;  
    yn2 = yn1; yn1 = yn;  
    return (yn);  
}
```

How does such a filter work ?

- Visualizing spectrograms and listening to some signal ...

#demo_filtering

- Ok, seems that the filter accentuates frequencies at around 5000 Hz and attenuates around 11000, but we'd like to know exactly => we want **frequency response** (*frekvenční / kmitočtová charakteristika*)
- Also, we would like to know if the filter will behave reasonably, i.e. not produce (for reasonable input) $+\infty$ or $-\infty$, that would lead to a destruction of our (expensive) audio equipment => we want to investigate **stability** (*stabilita*)

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Filter in the time domain

- Difference equation or convolution with impulse response
 - for FIR filters (working just with the input signal) – the same.
 - For IIR filters – convolution not doable (impulse response is infinite)

$$y[n] = b_0x[n] + b_1x[n-1] + \dots + b_Qx[n-Q] - a_1y[n-1] - \dots - a_Py[n-P]$$

$$y[n] = \sum_{k=0}^Q b_k x[n-k] - \sum_{k=1}^P a_k y[n-k]$$

$$y[n] = h[n] \star x[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Filter in the frequency domain

- Let's make sure we know which frequencies we'll be working with – **normalized angular frequency** (normovaná kruhová / úhlová frekvence) in **radians**
 - 0 Hz corresponds to 0 rad
 - $F_s / 2$ corresponds to π rad
 - F_s corresponds to 2π rad.
- Obtaining spectra of input and output signals by Discrete time TFT (Fourierova transformace s diskretním časem)
$$\omega = 2\pi \frac{f}{F_s}$$
- Practically, always computed by **DFT / FFT** !

$$X(e^{j\omega}) = DTFT \{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

$$Y(e^{j\omega}) = DTFT \{y[n]\} = \sum_{n=-\infty}^{\infty} y[n]e^{-j\omega n}$$

For one single frequency $\omega_1 \dots$

A complex exponential or cosine have only one frequency:

- Transfer or transfer coefficient (**přenos nebo činitel přenosu**) – complex number $H(e^{j\omega_1})$
- Its magnitude determines **how much** the signal will change $|H(e^{j\omega_1})|$
- Its phase determines how much the signal gets **shift** $\arg H(e^{j\omega_1})$

However, we don't want just a single frequency, **we want them all !**

Frequency response (kmitočtová / frekvenční charakteristika)

- Determines the behavior of a system on all frequencies
- **Complex** function of frequency: $H(e^{j\omega})$
- **Magnitude frequency response** (amplitudová / modulová kmitočtová / frekvenční charakteristika) is the most crucial – which frequencies will be attenuated $|H(e^{j\omega})|$ or amplified?
- **Phase frequency response** (fázová / argumentová kmitočtová / frekvenční charakteristika) is sometimes important as well (think of Hi-Fi app $\arg H(e^{j\omega})$)

Relation of time and frequency

- Time: **convolution** $y[n] = h[n] \star x[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$

- Frequency: **mutlification** $Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$

- Remember we can obtain the spectra of input and output by DTFT:

$$X(e^{j\omega}) = DTFT \{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad Y(e^{j\omega}) = DTFT \{y[n]\} = \sum_{n=-\infty}^{\infty} y[n]e^{-j\omega n}$$

- So that it seems that we can obtain the frequency response by DTFT of the impulse response and we're done

$$H(e^{j\omega}) = DTFT \{h[n]\} = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n}$$

Frequency response by DTFT of impulse response

- For FIR filters (**finite** impulse response) – perfectly **DOABLE** – DTFT (computed by DFT) of $h[n]$.
- Remember:
 - $h[k] = b_k$ for $k = 0 \dots Q$
 - $h[k] = 0$ otherwise
- For IIR (infinite impulse response) – **NOT DOABLE** – we can't DFT infinite sequence of numbers !

#iir_response

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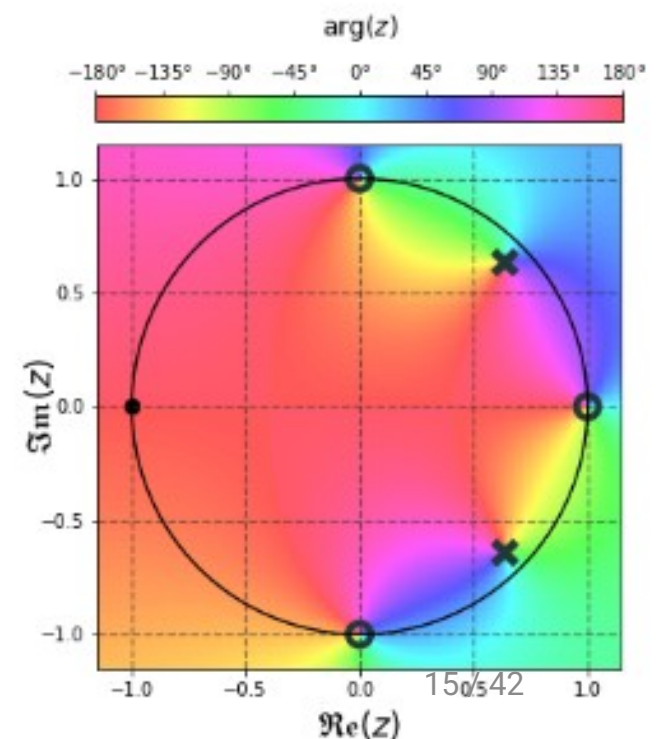
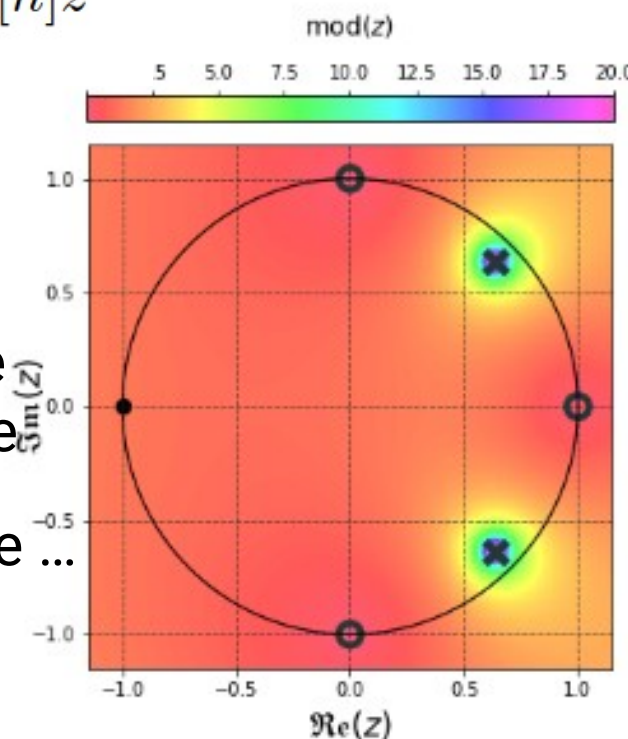
z-transform (z-transformance)

- z is a complex variable defined over the whole complex plane
- z-transform maps discrete real signal to complex function over complex plane.

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

- How to write it :

- How to $x[n] \xrightarrow{z} X(z)$
 - As “mountains” over a surface
 - Attention, these mountains are complex, so we need to visualize magnitude and phase ...



Properties of z-transform

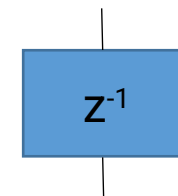
- For signal $x[n]$: $x[n] \xrightarrow{\mathcal{Z}} X(z)$
- When we multiply with a constant: $ax[n] \xrightarrow{\mathcal{Z}} aX(z)$
- Linear combination of 2 (or more) signals: $a_1x_1[n] + a_2x_2[n] \xrightarrow{\mathcal{Z}} a_1X_1(z) + a_2X_2(z)$
- Delay of the signal:

$$x[n - k] \xrightarrow{\mathcal{Z}} X(z)z^{-k}$$

$$\sum_{n=-\infty}^{\infty} x[n - k]z^{-n} = \sum_{n=-\infty}^{\infty} x[n]z^{-(n+k)} = \sum_{n=-\infty}^{\infty} x[n]z^{-n}z^{-k} = z^{-k} \sum_{n=-\infty}^{\infty} x[n]z^{-n} = z^{-k}X(z)$$

- That's why we mark the delay boxes in the sch
 - This box has **one** input and **one** output, not more !!!!!....

$$x[n - 1] \xrightarrow{\mathcal{Z}} X(z)z^{-1}$$



Relation of z-transform and DTFT

- Just write them next to each other ... they look very similar

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

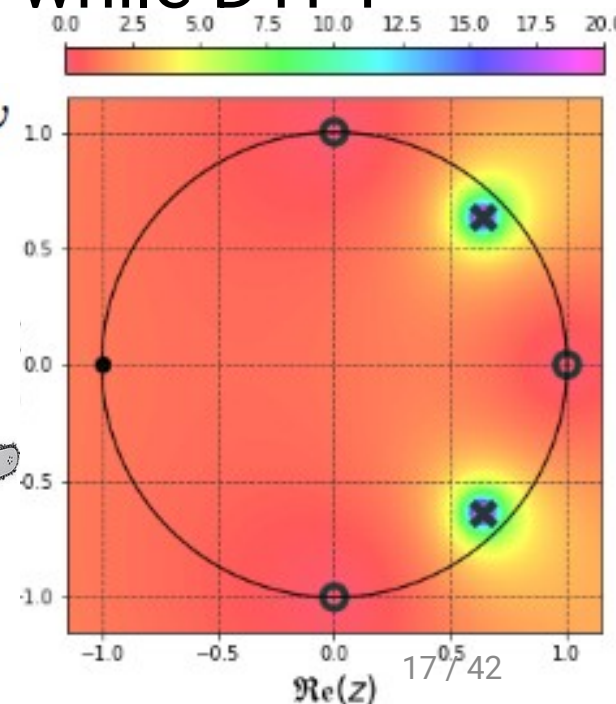
- z-transform works over the whole complex plane, while DTFT works only for values $e^{j\omega}$.

- Converting z-transform into DTFT is sim. $z \longrightarrow e^{j\omega}$

- Or (in fancy mathematical writing)

$$X(e^{j\omega}) = X(z)|_{z=e^{j\omega}}$$

- Imagine this by taking a machine saw and cutting around the unit circle!



Letting z-transform process the difference equation

$$y[n] = b_0x[n] + b_1x[n-1] + \dots + b_Qx[n-Q] - a_1y[n-1] - \dots - a_Py[n-P]$$

- Remember the z-rules:

- See a signal ? Rewrite it as its z-transform.
- See a constant ? Copy-paste
- Is the signal delayed by k samples ? Mutliply its z-transform by z^{-k}

$$Y(z) = b_0X(z) + b_1X(z)z^{-1} + \dots + b_QX(z)z^{-Q} - a_1Y(z)z^{-1} - \dots - a_PY(z)z^{-P}$$

- Our goal is to re-arrange it to obtain fraction

- Why ?

$$\frac{Y(z)}{X(z)}$$

Transfer function (přenosová / systémová funkce) $H(z)$

- z-representation of the output over z-representation of the input $H(z) = \frac{Y(z)}{X(z)}$
- Output / input is a very common way to express system's behavior
 - Amplification / attenuation = output voltage / input voltage
 - Currency exchange rate = target currency amount / source currency amount
 - Return c $Y(z) + a_1 Y(z)z^{-1} + \dots + a_p Y(z)z^{-P} = b_0 X(z) + b_1 X(z)z^{-1} + \dots + b_Q X(z)z^{-Q}$
- For our filter:

$$Y(z)(1 + a_1 z^{-1} + \dots + a_p z^{-P}) = X(z)(b_0 + b_1 z^{-1} + \dots + b_Q z^{-Q})$$

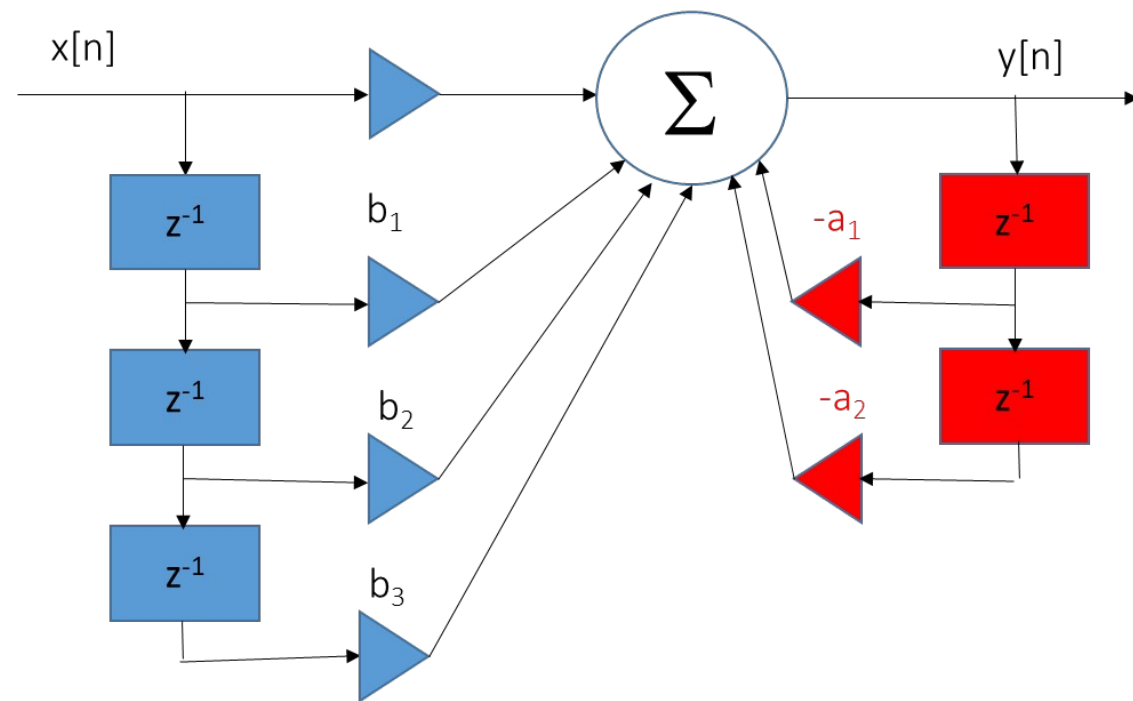
$$\frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + \dots + b_Q z^{-Q}}{1 + a_1 z^{-1} + \dots + a_p z^{-P}}$$

Transfer function II

- Another way to represent the filter:

$$H(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_Q z^{-Q}}{1 + a_1 z^{-1} + \dots + a_P z^{-P}} = \frac{\sum_{k=0}^Q b_k z^{-k}}{1 + \sum_{k=1}^P a_k z^{-k}}$$

- Two **polynomials** (**polynomy**)
 - Numerator, fully determined by coefficients multiplying the inputs of filter
 - Denominator, fully determined by coefficients multiplying the outputs of filter
 - We define $a_0 = 1$



$$B(z) = \sum_{k=0}^Q b_k z^{-k}$$

$$A(z) = 1 + \sum_{k=1}^P a_k z^{-k}$$

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From transfer function to frequency response

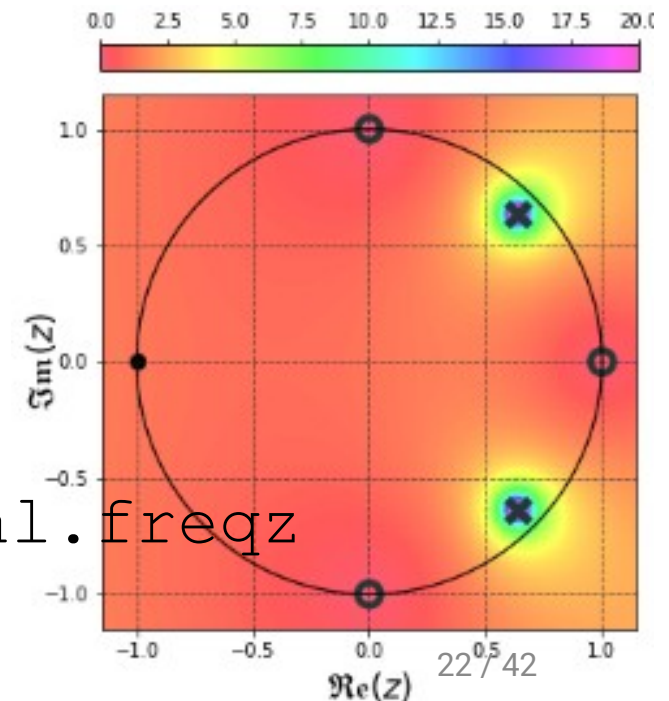
- We have $H(z)$ but we want $H(e^{j\omega})$
- Fortunately, there is this relation of z-transform and DTFT ...

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

- So that we can replace $z \rightarrow e^{j\omega}$
- this means taking the machine saw and cutting around the unit circle.
- Mathematically:

$$H(e^{j\omega}) = H(z)|_{z=e^{j\omega}} = \frac{\sum_{k=0}^Q b_k e^{-j\omega k}}{1 + \sum_{k=1}^P a_k e^{-j\omega k}}$$

- Numerator and denominator coefficients ! Fortunately, we have `scipy.signal.freqz`



How to visualize the frequency response ?

- Run only from $\omega = 0$ rad (corresponds to 0 Hz) till $\omega = \pi$ rad (corresponds to $F_s / 2$ Hz).
 - Select a reasonable number of points (256 is good)
 - It is a complex function of frequency, so visualize
 - Magnitude: often in deciBells as
 - Phase.
- $$H_{dB}(e^{j\omega}) = 20 \log_{10} |H(e^{j\omega})| = 10 \log_{10} |H(e^{j\omega})|^2$$
- `scipy.signal.freqs(B, A)` : **B** is vector of **numerator coefficients**, **A** vector of **denominator coefficients** (including $a_0 = 1$).
 - In case we need frequencies in Hz, just divide ω by 2π and denormalize by multiplying by F_s

#freqz

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Factorizing $B(z)$ and $A(z)$

- Let's massage the fraction to have polynomials with positive powers of z :

$$H(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_Q z^{-Q}}{1 + a_1 z^{-1} + \dots + a_P z^{-P}} = \frac{z^{-Q}(b_0 z^Q + b_1 z^{Q-1} + \dots + b_Q)}{z^{-P}(z^P + a_1 z^{P-1} + \dots + a_P)}$$

- We can now **factorize** (*rozložit*) the blue and red polynomials.
- In case we find **roots** of a polynomial (*kořeny polynomu*),

$$x^2 + bx + c = 0 \longrightarrow x_1, x_2$$

we can write it in a **factorized form**.

$$x^2 + bx + c = (x - x_1)(x - x_2)$$

Exercises of factorization

Check results of these exercises with **#roots** obtained by `np.roots`

- Sometimes, we remember some decomposition formula and we don't need much work

- For others, like $x^2 + 2x + 1 = (x + 1)(x + 1) = (x - (-1))(x - (-1)) \rightarrow x_1 = -1, x_2 = -1$ we need to solve quadratic equation – we're not at high school, we **can** do square root of a negative number !

- For higher order polynomial, roots are **real** or **complex conjugate pairs**.

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{2 \pm \sqrt{4 - 8}}{2} = \frac{2 \pm 2j}{2}$$

$$x_1 = 1 + j, \quad x_2 = 1 - j$$

Factorizing the transfer function

$$H(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_Q z^{-Q}}{1 + a_1 z^{-1} + \dots + a_P z^{-P}} = \frac{z^{-Q}(b_0 z^Q + b_1 z^{Q-1} + \dots + b_Q)}{z^{-P}(z^P + a_1 z^{P-1} + \dots + a_P)}$$

$$= b_0 z^{P-Q} \frac{(z - n_1)(z - n_2) \dots (z - n_Q)}{(z - p_1)(z - p_2) \dots (z - p_P)}$$

Or, in fancy mathematical writing with $\prod$ denoting product:

$$H(z) = b_0 z^{P-Q} \frac{\prod_{k=1}^Q (z - n_k)}{\prod_{k=1}^P (z - p_k)}$$

Let's take an example of our

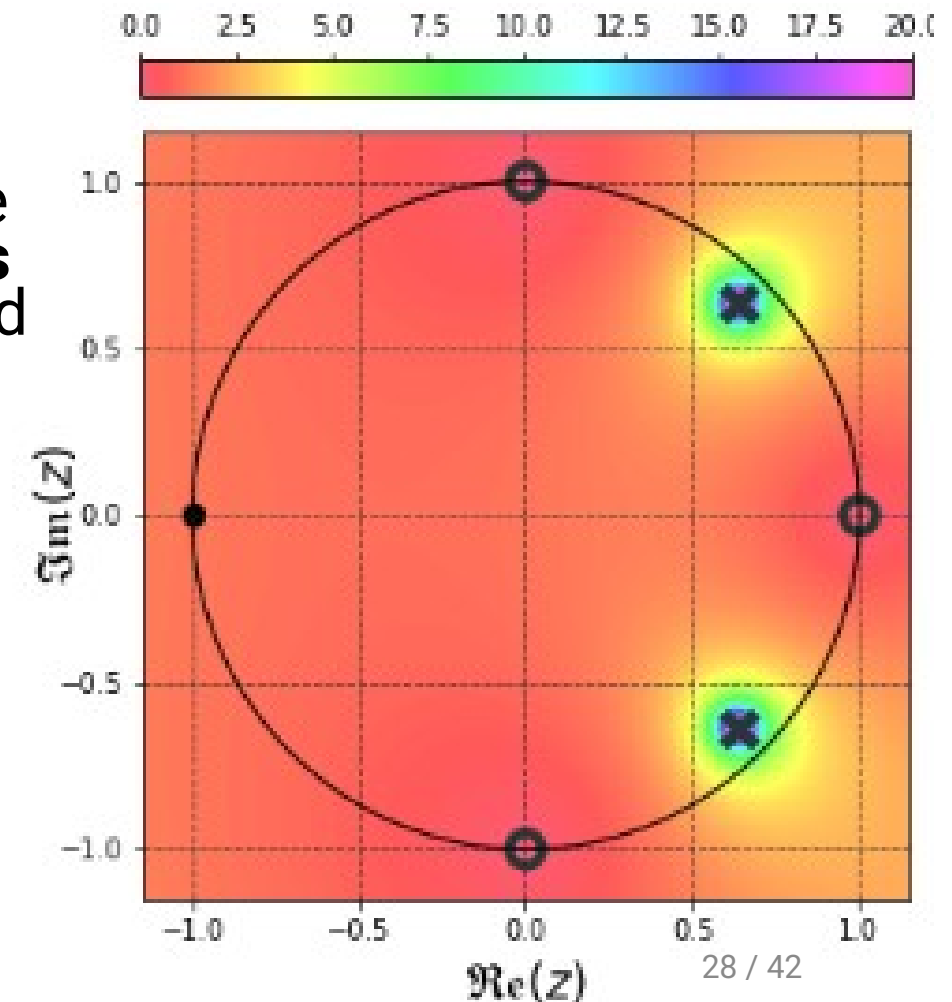
- $b_0 = 1, b_1 = -1, b_2 = 1, b_3 = -1$

- $a_0 = 1, a_1 = -1.2723, a_2 = 0.81 \rightarrow$ roots $p_1 = 0.9e^{j\pi/4}, p_2 = 0.9e^{-j\pi/4}$

Understanding numerator and denominator

$$b_0 z^{P-Q} \frac{(z - n_1)(z - n_2) \dots (z - n_Q)}{(z - p_1)(z - p_2) \dots (z - p_P)}$$

- Let's look at the behavior of complex function $H(z)$ at the roots.
- In case z falls into a root of the numerator, $z = n_1, z = n_2$, etc., the whole function $H(z)$ will be **zero**. The roots are therefore called **zero points** or simply **zeros** (*nulové body / nuly*) and marked **O**.
- In case z falls into a root of the denominator $z = p_1, z = p_2$, etc., the whole function $H(z)$ will be **infinity**. The roots are therefore called **poles** (*póly*) and marked **X**.

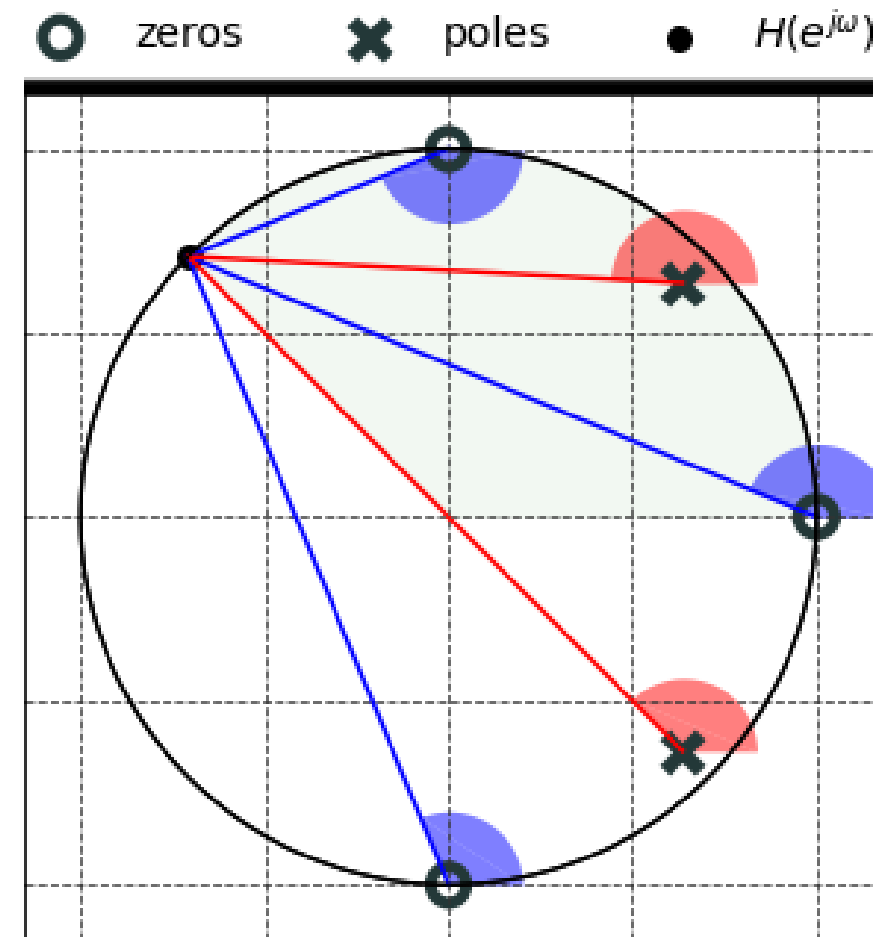


From zeros and poles to frequency response

- With the factorization, frequency response can be written as

$$H(e^{j\omega}) = b_0 e^{j\omega(P-Q)} \frac{(e^{j\omega} - n_1)(e^{j\omega} - n_2) \dots (e^{j\omega} - n_Q)}{(e^{j\omega} - p_1)(e^{j\omega} - p_2) \dots (e^{j\omega} - p_P)}$$

- We can imagine
 - $e^{j\omega}$ as a point on the unit circle. Its position corresponds to the frequency.
 - Each **blue bracket** as a vector starting in a zero point and going to $e^{j\omega}$
 - Each **red bracket** as a vector starting in a pole and going to $e^{j\omega}$



From zeros and poles to frequency response II

$$H(e^{j\omega}) = b_0 e^{j\omega(P-Q)} \frac{(e^{j\omega} - n_1)(e^{j\omega} - n_2) \dots (e^{j\omega} - n_Q)}{(e^{j\omega} - p_1)(e^{j\omega} - p_2) \dots (e^{j\omega} - p_P)}$$

Each bracket is just a complex number, so the whole thing is just a multiplication and division of complex numbers. Remember:

- For multiplication, we multiply magnitudes and add phases

$$z_1 z_2 = r_1 r_2 e^{j(\phi_1 + \phi_2)}$$

- For division, we divide magnitudes and subtract phases

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{j(\phi_1 - \phi_2)}$$

- So

- When computing the magnitude of frequency response, take into account only the magnitudes (lengths of blue and red vectors)
- When computing the phase of the frequency response, take into account only the phases (angles of blue and red vectors).

From zeros and poles to frequency response III

$$H(e^{j\omega}) = b_0 e^{j\omega(P-Q)} \frac{(e^{j\omega} - n_1)(e^{j\omega} - n_2) \dots (e^{j\omega} - n_Q)}{(e^{j\omega} - p_1)(e^{j\omega} - p_2) \dots (e^{j\omega} - p_P)}$$

- Magnitude (need to take into account b_0 if it is not 1):

$$|H(e^{j\omega})| = b_0 \frac{|e^{j\omega} - n_1| |e^{j\omega} - n_2| \dots |e^{j\omega} - n_Q|}{|e^{j\omega} - p_1| |e^{j\omega} - p_2| \dots |e^{j\omega} - p_P|} = b_0 \frac{\text{product of lengths of blue vectors}}{\text{product of lengths of red vectors}}$$

- Phase (need to take into account $e^{j\omega(P-Q)}$ if P and Q are not the same): @@@spatne opravit !

$$\begin{aligned} \arg H(e^{j\omega}) &= (P - Q)\omega \frac{\arg(e^{j\omega} - n_1) \arg(e^{j\omega} - n_2) \dots \arg(e^{j\omega} - n_Q)}{\arg(e^{j\omega} - p_1) \arg(e^{j\omega} - p_2) \dots \arg(e^{j\omega} - p_P)} \\ &= (P - Q)\omega + \text{sum of angles of blue vectors} - \text{sum of angles of red vectors} \end{aligned}$$

Exercise on our filter for 4 typical frequencies

- Only magnitudes, but take a time and play also with phases !

#zeros_poles_to_freq_response

- $\omega = 0$
$$|H(e^{j\omega})| = 1 \frac{|1 - 1| |1 - (-j)| |1 - j|}{|1 - 0.9e^{j\pi/4}| |1 - 0.9e^{-j\pi/4}|} = 0$$
- $\omega = \pi / 4$
$$|H(e^{j\omega})| = 1 \frac{|e^{j\pi/4} - 1| |e^{j\pi/4} - (-j)| |e^{j\pi/4} - j|}{|e^{j\pi/4} - 0.9e^{j\pi/4}| |e^{j\pi/4} - 0.9e^{-j\pi/4}|} = \frac{0.75 \ 0.75 \ 1.8}{0.1 \ 1.4} = 7.3$$
- $\omega = \pi / 2$
$$|H(e^{j\omega})| = 1 \frac{|j - 1| |j - (-j)| |j - j|}{|j - 0.9e^{-j\pi/4}| |j - 0.9e^{-j\pi/4}|} = 0$$
- $\omega = \pi$
$$|H(e^{j\omega})| = 1 \frac{|-1 - 1| |-1 - (-j)| |-1 - j|}{|-1 - 0.9e^{-j\pi/4}| |-1 - 0.9e^{-j\pi/4}|} = \frac{\sqrt{2} \ 2 \ \sqrt{2}}{1.75 \ 1.75} = 1.84$$

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What is stability

- Bounded input $\rightarrow$ bounded output.
- If input $x[n]$ is in interval $[-C, C]$, we can find D such that the output $y[n]$ is bounded in interval $[-D, D]$
 - C is usually 1, as we are normalizing the input signal.
- Unstable filter usually produces values $+\infty$ or $-\infty$ - you probably witnessed (for example at a rock concert) what it means, if a system becomes unstable.

Stability of FIR filters

- Filtering works with the difference equation processing only input samples (coefficients b_k), which is equivalent to convolution with impulse response (remember: for FIR filters, impulse response = coefficients)

$$y[n] = b_0x[n] + b_1x[n-1] + b_2x[n-2] + b_3x[n-3]$$

$$y[n] = h[0]x[n] + h[1]x[n-1] + h[2]x[n-2] + h[3]x[n-3]$$

- Example for $C = 1$ (input signal normalized to +/-1) and 4 coefficients:
 $b_0 = 1.5, b_1 = 1, b_2 = 0.5, b_3 = -1.$

- the worst case happens if input samples have maximum magnitude and their sign matches the sign of the coefficient:

$$y_{max}[n] = 1.5 \times 1 + 1 \times 1 + 0.5 \times 1 + (-1) \times (-1) = 4$$

Stability of FIR filters II

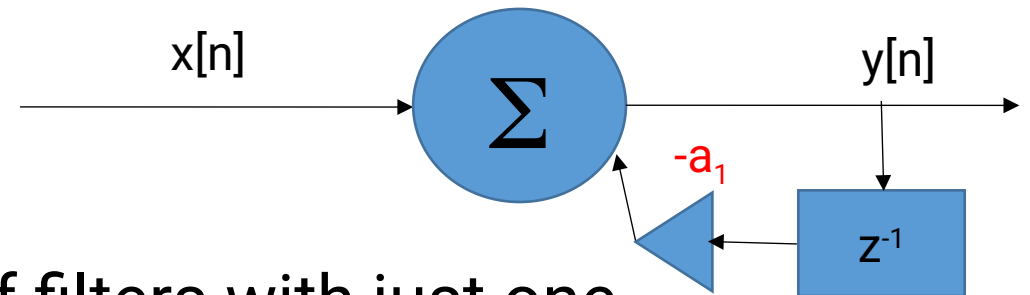
- Generally:

$$D = \sum_{k=0}^Q C |b_k|$$

- We can find output limitation D for any input limitation C
=> FIR filter is always **stable**

Stability of IIR filters

- Sum of impulse response does not work – it is infinite ☹
- We can determine stability only for the simplest filter
 $|a_1| < 1$
- But with the factorization



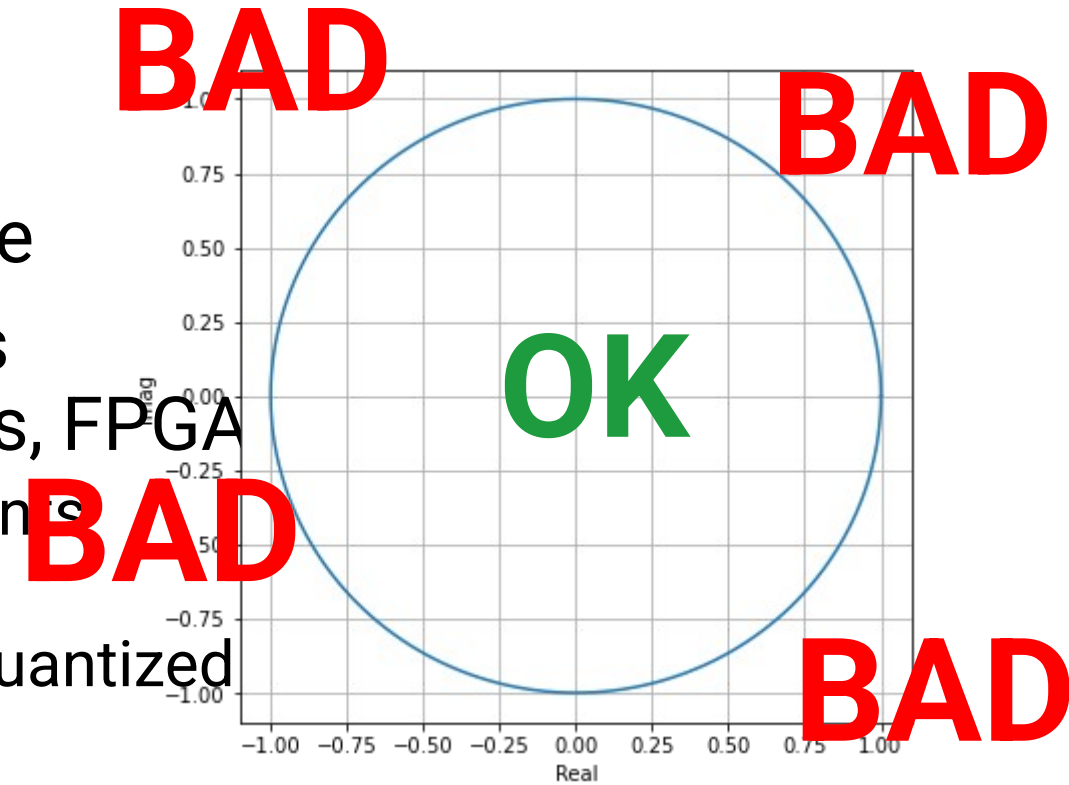
we can write the transfer function of filters with just one coefficient – the poles are the coefficients ! Therefore, the condition is

$$b_0 z^{P-Q} \frac{(z - n_1)(z - n_2) \dots (z - n_Q)}{(z - p_1)(z - p_2) \dots (z - p_P)}$$

$$|p_k| < 1$$

Stability of IIR filters II

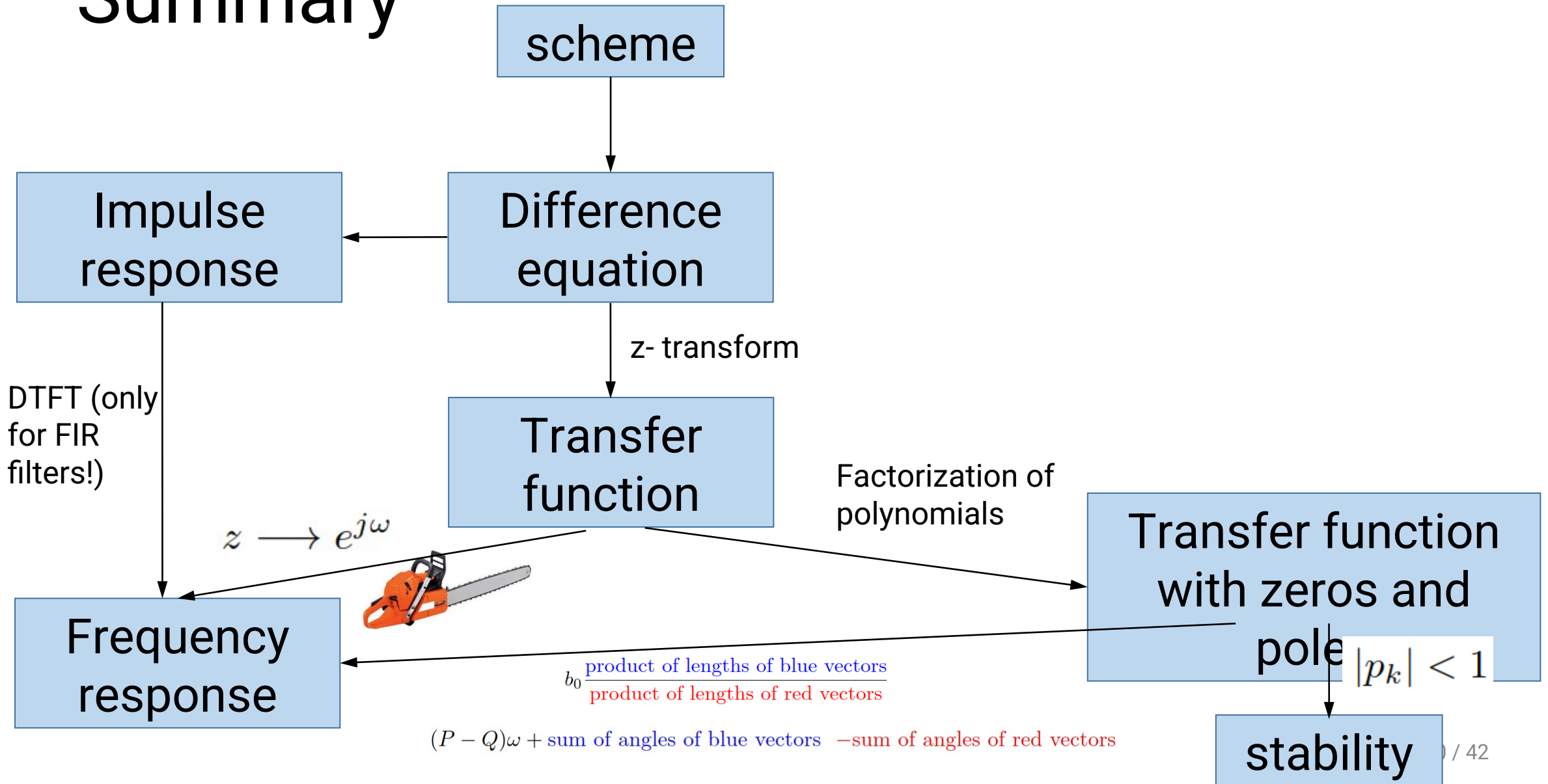
- All poles must be inside the unit circle
- Beware when implementing IIR filters with less precision (signal processors, FPGA)
 - Even if your filter is stable with coefficients expressed as floats / doubles
 - It can become unstable once they are quantized to less bits !
- Check more advanced signal processing courses, for example CZSa.



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Summary II

As usual, be nice with **visualization** of your results:

- Ask your colleague / boss / customer, how he/she wants the frequency axis:
 - For normalized angular frequencies, go from $\omega = 0$ rad to $\omega = \pi$ rad
 - For regular frequencies, go from 0 Hz to $F_s / 2$ Hz.
- Pick good number of points.
- Show the result in linear or log (decibel) scale.
- Phase is not always requested, but be ready to provide it (`np.angle`)

Summary III

- Usual filter design functions produce stable filters
- But better check it, especially when trying something own and/or special: poles must be inside unit circle.
- Be careful when designing very sharp filters, their poles tend to be deadly close to the unit circle !
- When modifying filter coefficients (representation with less bits), check again, quantization can kick the poles out of the unit circle.